

Alex (Rui) Gan

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Education

University of Toronto - St. George

Honours Bachelor of Science

Computer Science Specialist, Focus in Artificial Intelligence

Sep 2018 – Sep 2022

Experience

Amazon

Toronto, ON

Software Development Engineer II

Aug 2022 – Present

- Designed and built Ad Pricing **Skill** on agentic AI platform with secure cross-service data access across **4 auth systems** (Cognito, IAM, Redshift, Lake Formation), reducing pricing effort from **2 hours to 3 minutes** per deal for business development teams, enabling the global launch of Spotlight SOV with **\$50MM ARR** target.
- Drove development of the Books Ads **Agentic AI Platform** built on **MCP** open standard, **RAG**, and AWS Bedrock — including AI agent skills, access control, and usage metrics — cutting priority oncall ticket resolution time by **4x**, reducing advertiser insights analysis time from 2 days to minutes, increasing AE productivity by **20-30x** and driving **\$6MM annual revenue**.
- Integrated a FlanT5 transformer-based semantic recommender model into SP Keyword Recommendation System across **7 marketplaces**. Built Airflow-orchestrated Spark ETL data pipeline processing **35B** keyword-ASIN pairs daily into DynamoDB. A/B experiment passed launch guardrails with stat-sig revenue lift in US/CA/UK improving keyword coverage 45% → **80%** with **\$3MM** projected ARR.
- Led end-to-end development of **Amazon Posts and Post Boost**, coordinating **11+ teams** to enable one-click conversion of author content into Sponsored Brands ads. Reduced ad creation time by **70%**, with Boosted Post Ads increasing ROAS by 50%.
- Built automated oncall tool to backfill missing data and enable internal API calls, reducing customer issue resolution time by **10x** and reducing microservices integration effort by **60x**. Won organization's All Hands **Customer Obsession** award.

Amazon

Toronto, ON

Software Development Engineer Intern

May 2021 – Jul 2021

- Built operational excellence tool with **REST APIs** using **React** and **AWS** services (Lambda, API Gateway, SQS, CloudWatch, S3), enabling teams to build internal tools more efficiently.

Trinity College Chinese Students Association, University of Toronto

Toronto, ON

Web Developer

Jun 2020 – Apr 2021

- Developed club website for event information and member engagement to promote diversity.

Skills

Programming:	Python, Java, TypeScript, JavaScript, Scala, SQL, HTML/CSS, C/C++, C#
Frameworks & Tools:	Spring, React, Node.js, Express, Apache Spark, Docker, Git, Linux, MongoDB
Cloud:	AWS (Lambda, S3, DynamoDB, CloudWatch, CDK, ECS, Redshift, Glue, Athena)
AI/ML:	Bedrock AgentCore, Claude, LLM Orchestration, Prompt Engineering, MCP

Achievements and Awards

- Amazon Ads Org Customer Obsession Award
- School champion of the CEMC mathematics contests (Pascal and Gauss), scoring 142/150.